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Subject Name : Machine Learning

Unit No: 2

Unit Name : Supervised Learning: Regression and Classification

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Unit No.: 2.1

Faculty Name :

Regression vs. Classification

- Regression predicts continuous values (e.g., predicting a price).
- Classification categorizes data into discrete classes (e.g., spam vs. not-spam)

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Regression

- **Regression is a supervised learning technique used to predict continuous numerical values.**
- **Regression focuses on finding the mathematical relationship between input features (independent variables) and a target output (dependent variable) to estimate specific quantities like prices, temperatures, or sales.**
- It helps understand how changes in one or more factors influence a measurable outcome.
- It estimates the strength of relationships and predicts continuous outcomes (e.g., sales, prices) based on trends.
- **Purpose of Regression**: To forecast outcomes, identify key influencing factors, and map relationships between variables

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Core Concepts

- **Dependent Variable (y):** The target or outcome you want to predict (e.g., house price).
- **Independent Variables (X):** The input features used to make the prediction (e.g., square footage, location).
- **Best-Fit Line:** The goal of most regression algorithms is to find a line or curve that minimizes the distance (error) between the actual data points and the model's predictions.

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Common Types of Regression

1. **Simple Linear Regression:** Models the relationship between one independent variable and a target using a straight line.
2. **Multiple Linear Regression**: Uses two or more independent variables to predict a single outcome.
3. **Polynomial Regression:** Models non-linear relationships by using degree polynomial terms, creating a curved best-fit line.

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Common Types of Regression

4. **Regularized Regression (Ridge & Lasso):** Adds penalty terms to the model to prevent overfitting (when a model is too complex and fails on new data).
   - **Lasso (L1):** Can shrink some coefficients to zero, effectively performing feature selection.
   - **Ridge (L2):** Shrinks coefficients but keeps all features

5. **Logistic Regression:** Despite its name, it is primarily used for **classification** to predict the probability of a binary outcome (e.g., 0 or 1) using a sigmoid

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Real-World Applications of Regression

- **Application of Regression:** It is widely used in forecasting, risk analysis, decision-making and trend estimation
  - **Finance**: Forecasting stock prices, interest rates, and risk scores.
  - **Real Estate**: Estimating property values based on size, location, and amenities.
  - **Healthcare:** Predicting patient recovery times or identifying disease risk factors.
  - **Business**: Forecasting sales demand and optimizing marketing spend
  - What is the effect of price on sales? How this selling price affects sales in order to increase our market share.
  - Predict housing prices based on area?

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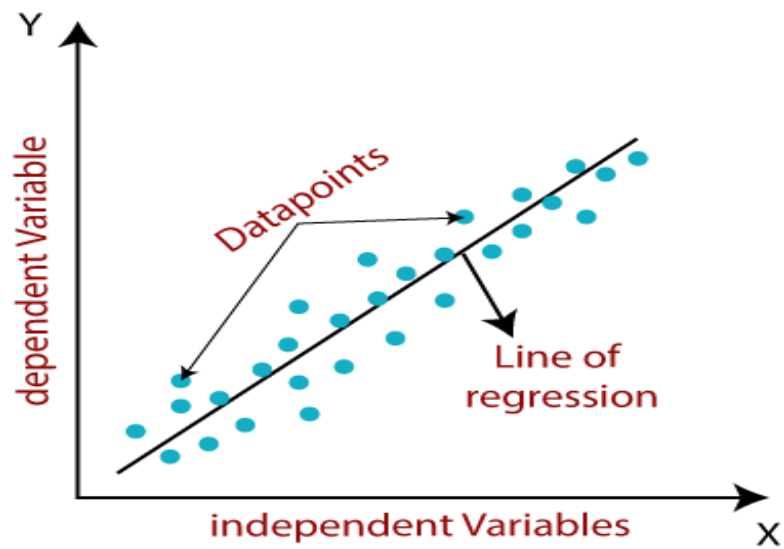
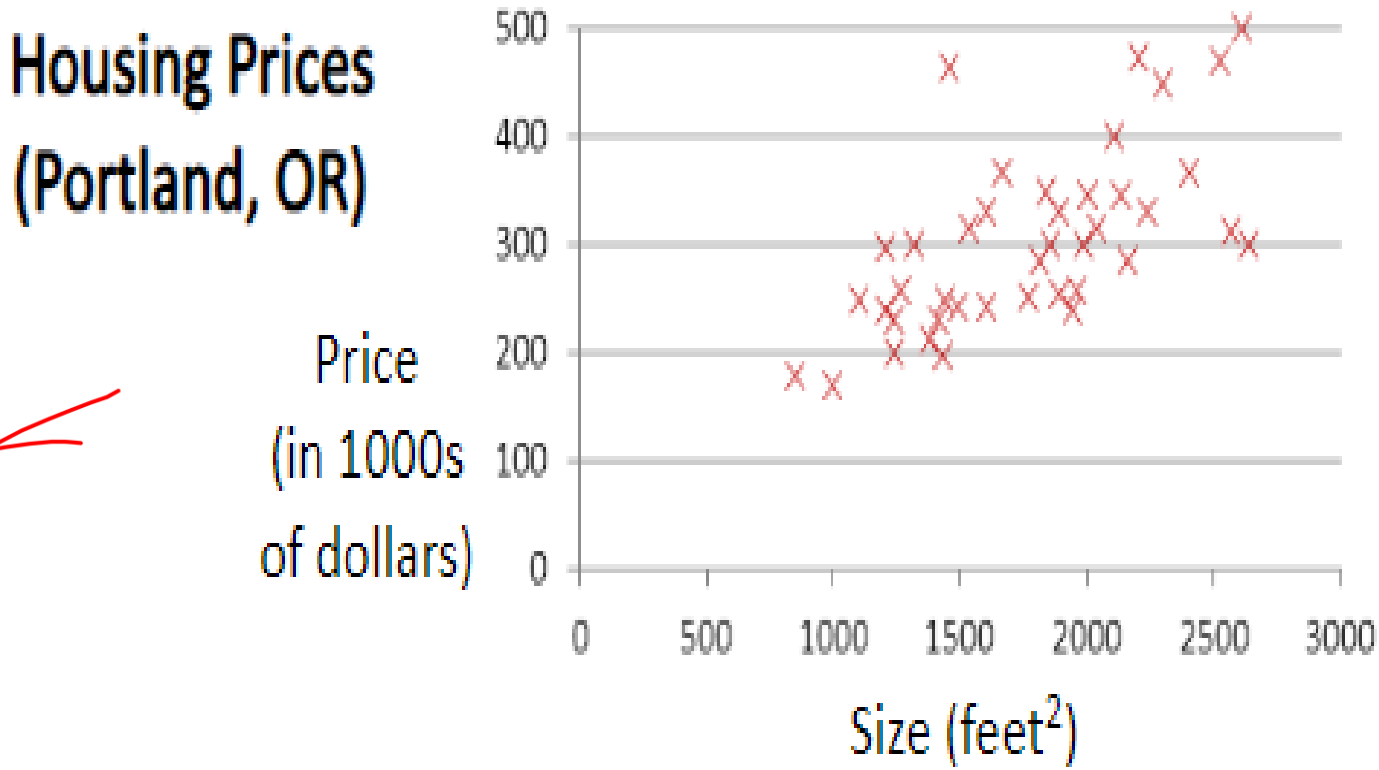
Simple Linear Regression (Univariate Linear Regression)

- Simple linear regression is a type of regression analysis where
  - number of independent variables and dependent variable is one
  - there is a linear relationship between the independent(x) and dependent(y) variable.
  - Also known as **univariate linear regression.**
- This method is mostly used for forecasting and finding out cause and effect relationship between variables.



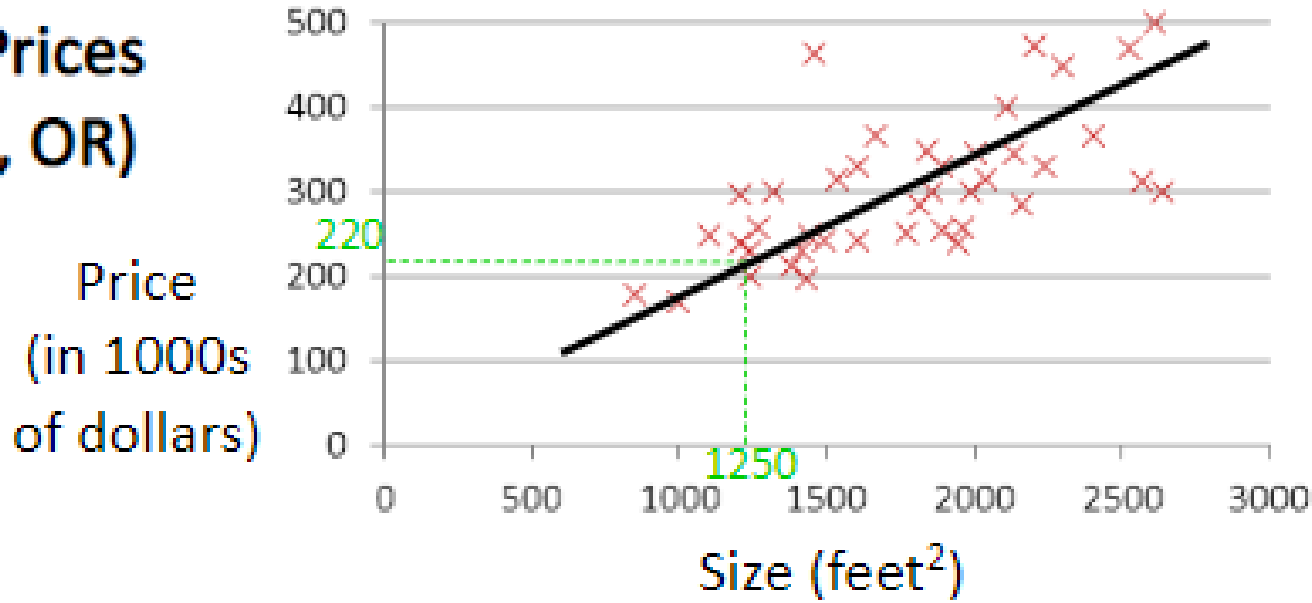
Simple Linear Regression



Which is the best straight line to fit the data?

Simple Linear Regression

Housing Prices (Portland, OR)



<u>Supervised Learning</u>

Given the "right answer" for each example in the data.

<u>Regression Problem</u>

Predict real-valued output

Classification : Discrete-valued output

Simple Linear Regression

Training set of housing prices (Portland, OR)

Size in feet^2 (x)	Price ($) in 1000's (y)
2104	460
1416	232
1534	315
852	178
…	…

m

Notation:

- **m** = Number of training examples
- **x**'s = "input" variable / features
- **y**'s = "output" variable / "target" variable

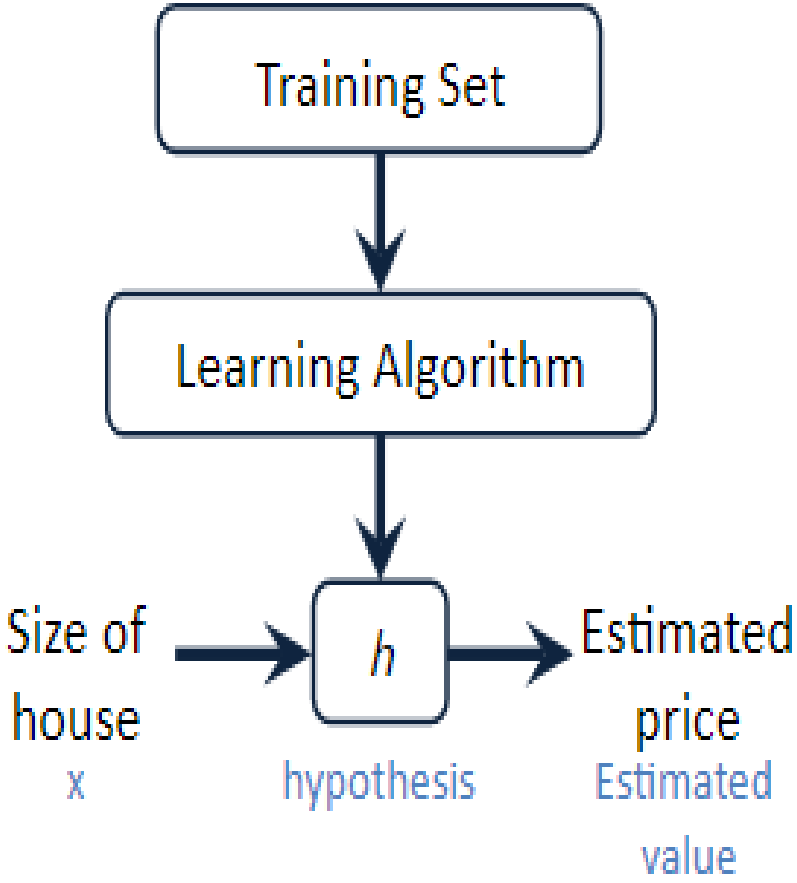
$x^{(1)} = 2104$

$x^{(2)} = 1416$

$y^{(1)} = 460$

- (x, y) – one training example
- $(x^{(i)}, y^{(i)})$ – ith trainingg example

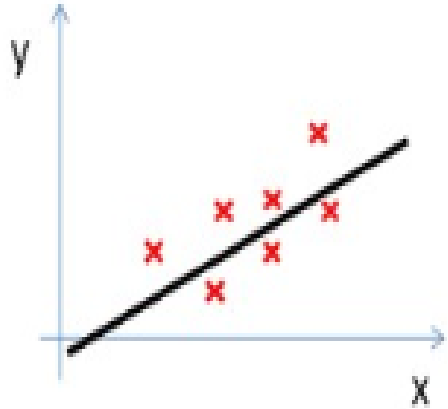
Simple Linear Regression



h maps from x's to y's

How do we represent *h*?

$h_\theta(x) = \theta_0 + \theta_1 x$



Linear regression with one variable.

Univariate linear regression.

One variable

Simple Linear Regression

- The expression $h(x) = \theta_1 x + \theta_0$ is the **hypothesis function** of **Simple Linear Regression**.
- Where:
  - $h(x)$ →Predicted output (also written as ŷ)
  - x→ Input feature (independent variable)
  - θ_1 →Slope (weight / coefficient)
  - θ_0 →Intercept (bias term)

- It represents a **straight line equation**: $y = mx + c$
- Where:
  - $\theta_1 = m$ (slope)
  - $\theta_0 = c$ (intercept)
- In **Simple Linear Regression**, we try to find the best values of θ_0 and θ_1
- So that the predicted value $h(x)$is as close as possible to the actual value y.

Simple Linear Regression

Training Set

Size in feet^2 (x)	Price ($) in 1000's (y)
2104	460
1416	232
1534	315
852	178
…	…

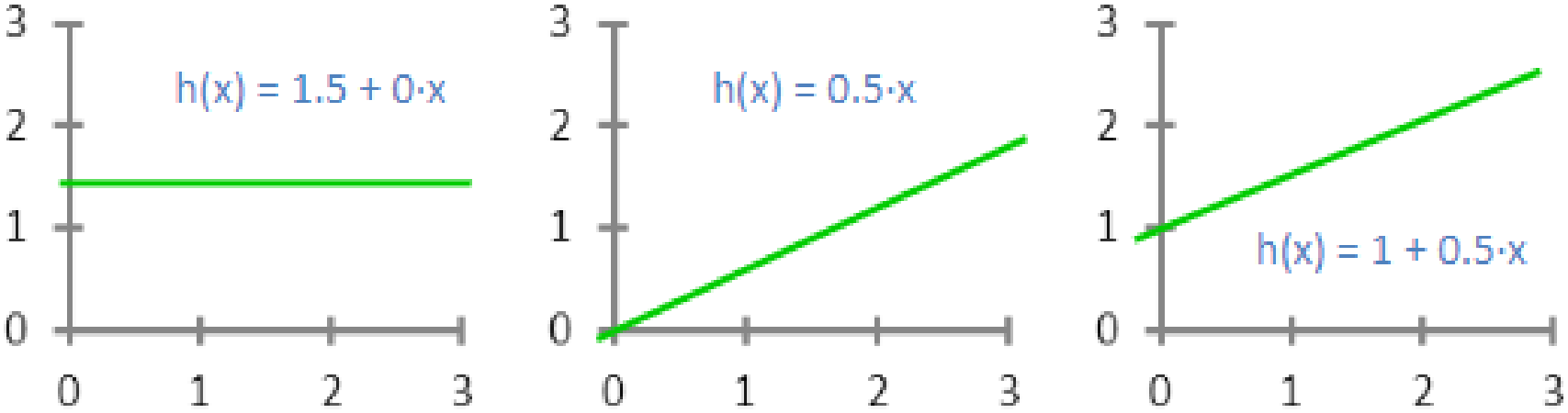
Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

θ_i's: Parameters

How to choose θ_i's ?

Impact of Parameter (θs)

$$h_\theta(x) = \theta_0 + \theta_1 x$$



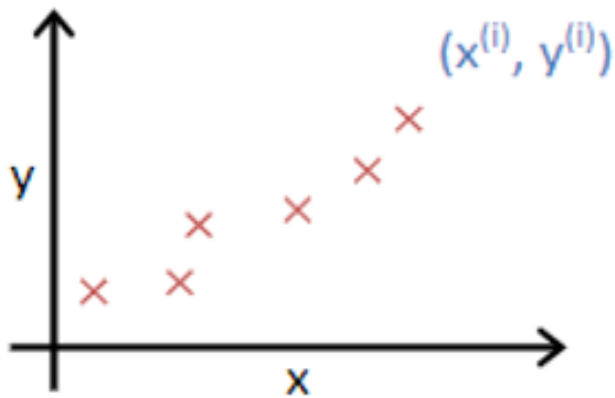
$\theta_0 = 1.5$
$\theta_1 = 0$

$\theta_0 = 0$
$\theta_1 = 0.5$

$\theta_0 = 1$
$\theta_1 = 0.5$

Simple Linear Regression: Cost Function

- **How to learn the values of the parameters θ_i ?**
- **Choose the parameters such that the prediction is close to the actual y-value for the training examples.**



$$\underset{\Theta_0, \Theta_1}{\text{minimize}} \frac{1}{2m} \sum_{i=1}^{m} \left(h_\Theta(x^{(i)}) - y^{(i)}\right)^2$$

$$h(x) = \Theta_0 + \Theta_1 x^{(i)}$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

Idea: Choose θ_0, θ_1 so that $h_\theta(x)$ is close to y for our training examples (x, y)

$\underset{\Theta_0, \Theta_1}{\text{Minimize}}\ J(\Theta_0, \Theta_1)$: Cost Function

Squared error function

Cost function is a measure of how close the predictions are to the actual y-values.

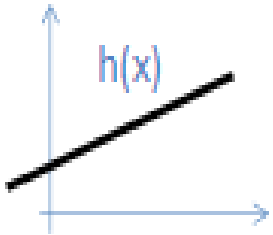
Simple Linear Regression Problem

Hypothesis:

$$h_\theta(x) = \theta_0 + \theta_1 x$$

Parameters:

$$\theta_0, \theta_1$$



Cost Function:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

Goal: $\underset{\theta_0, \theta_1}{\text{minimize}}\ J(\theta_0, \theta_1)$

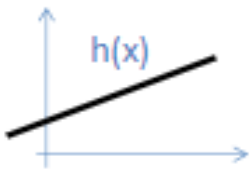
Simple Linear Regression: Cost Function

Hypothesis:

$$h_\theta(x) = \theta_0 + \theta_1 x$$

Parameters:

$$\theta_0, \theta_1$$



Cost Function:

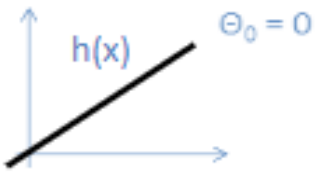
$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

Goal: $\underset{\theta_0, \theta_1}{\text{minimize}}\ J(\theta_0, \theta_1)$

Simplified

$$h_\theta(x) = \theta_1 x$$

$$\theta_1$$



$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

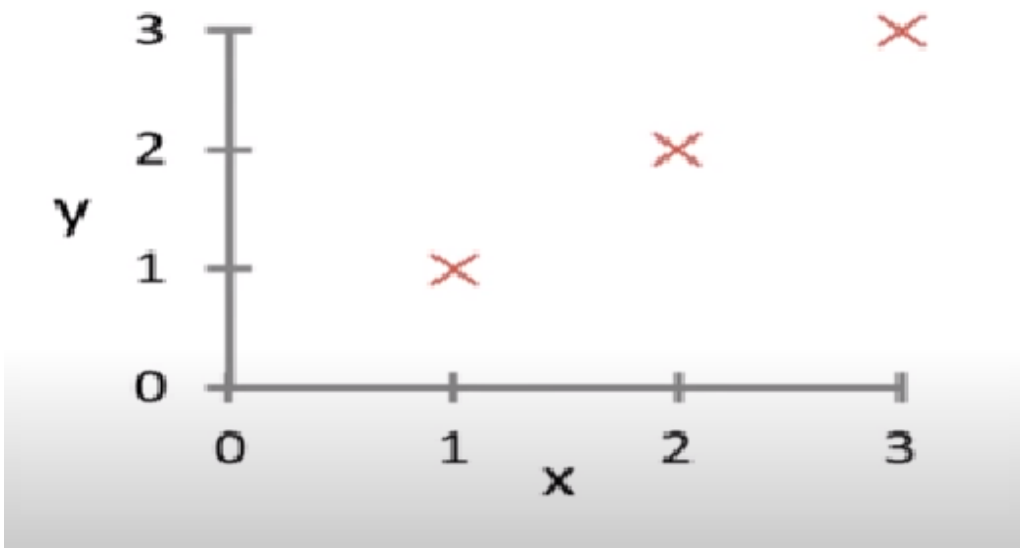
$\underset{\theta_1}{\text{minimize}}\ J(\theta_1)$

Simple Linear Regression: Cost Function for simplified model

Example: What would be my optimal value for θ_1 for the following example. We will try different values for θ_1.

$$h_\theta(x)$$

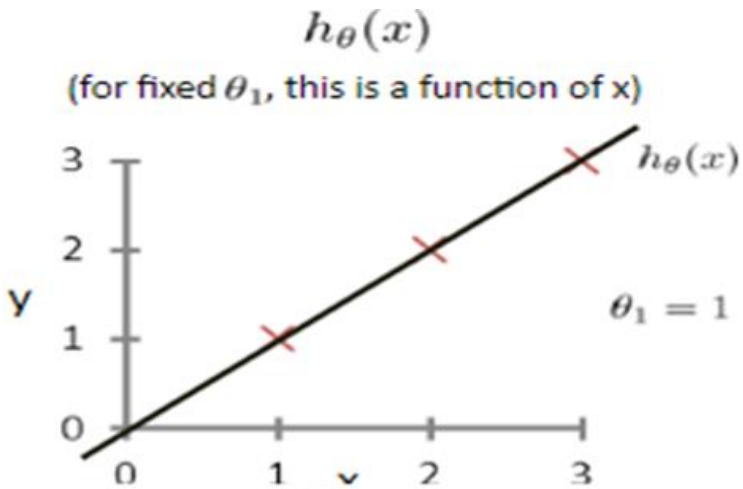
(for fixed θ_1, this is a function of x)



Ex1: $\theta_1 = 1$
Ex2: $\theta_1 = 0.5$
Ex3: $\theta_1 = 0$

Simple Linear Regression: Cost Function for simplified model

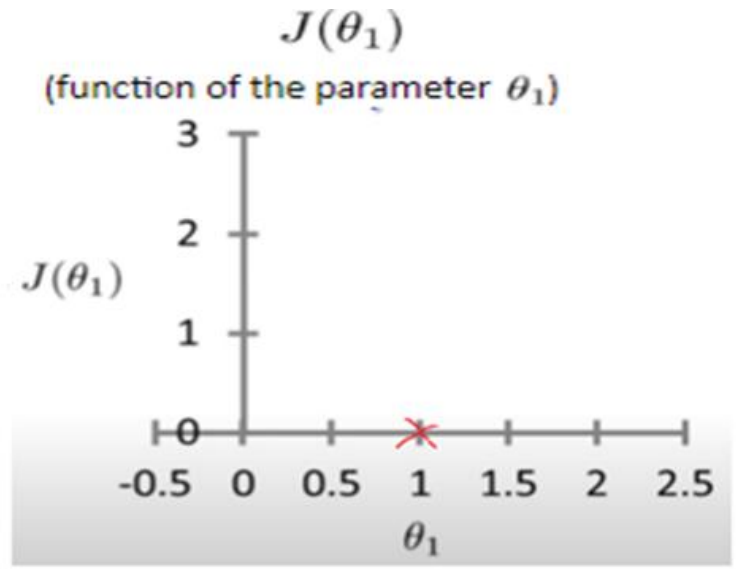
Ex 1: Find the value of $J(\theta_1)$ for $\theta_1 = 1$?



$$J(\theta_1) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

$$= \frac{1}{2m}\sum_{i=1}^{m}(\Theta_1 x^{(i)} - y^{(i)})^2$$

$$= \frac{1}{2m}(0^2 + 0^2 + 0^2)$$



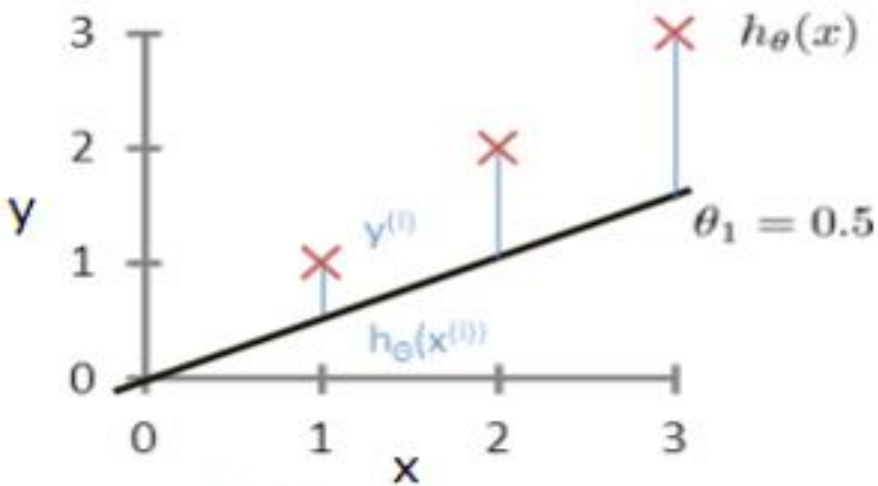
$J(1) = 0$

Simple Linear Regression: Cost Function

Ex2: Find the value of $J(\theta_1)$ for $\theta_1 = 0.5$?

$h_\theta(x)$

(for fixed θ_1, this is a function of x)

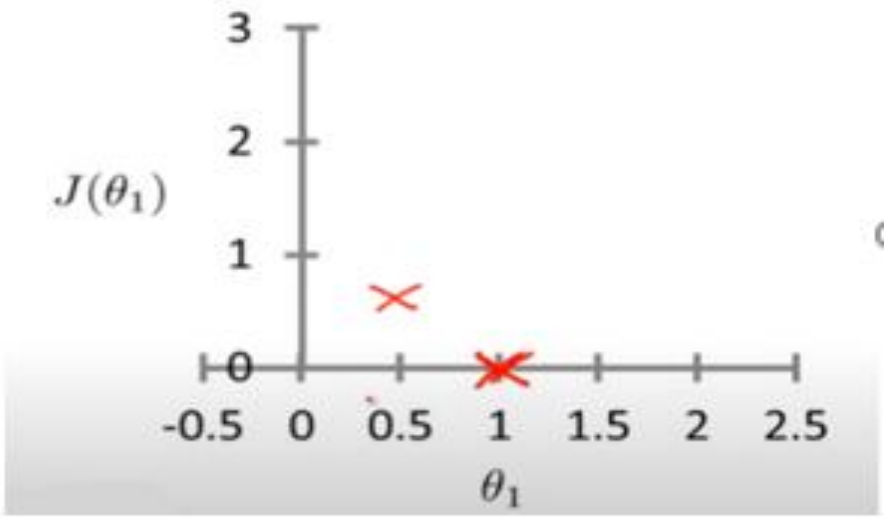


$$J(0.5) = \frac{1}{2\cdot 3}\sum_{i=1}^{3}\left[(0.5-1)^2 + (1-2)^2 + (1.5-3)^2\right]$$

$$= \frac{1}{6}\cdot(3.5) = 0.58$$

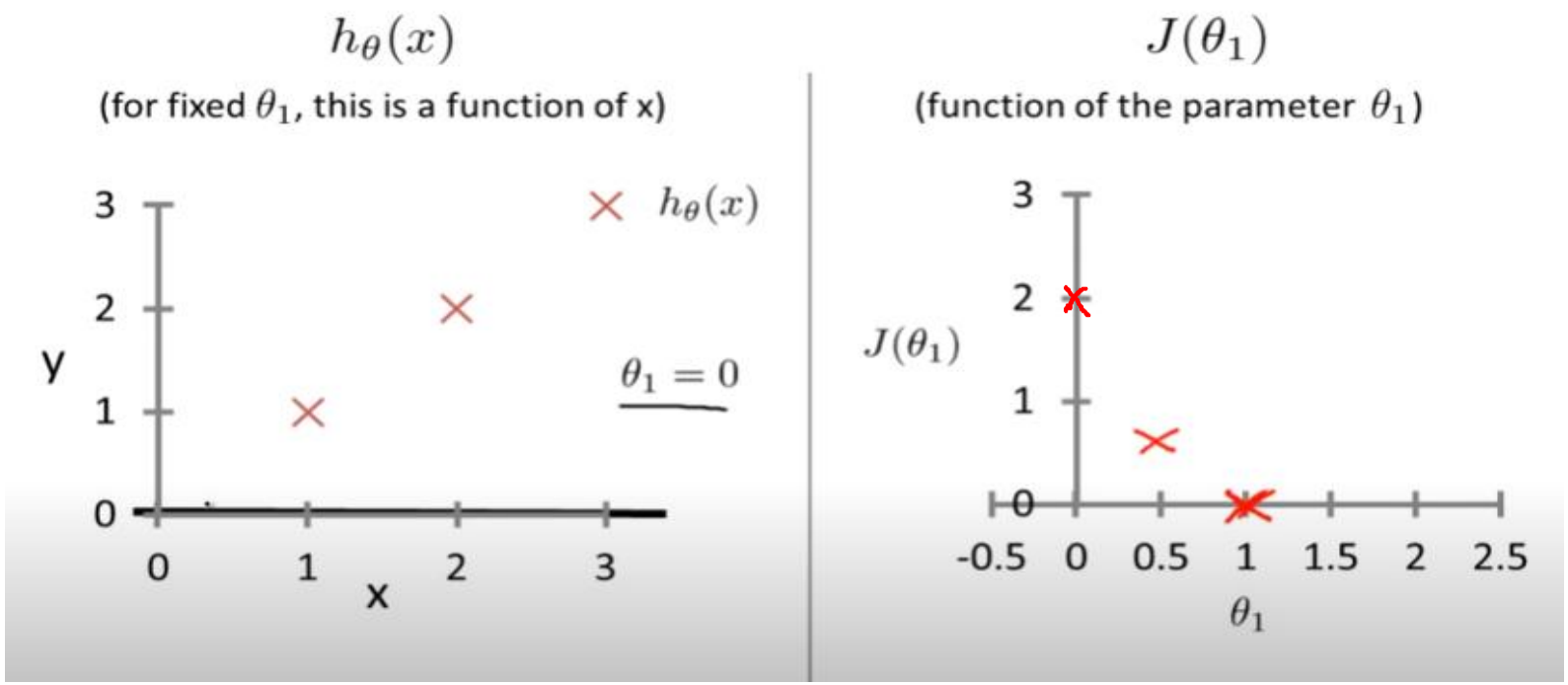
$J(\theta_1)$

(function of the parameter θ_1)



Simple Linear Regression: Cost Function

Ex3: Find the value of $J(\theta_1)$ for $\theta_1 = 0$?

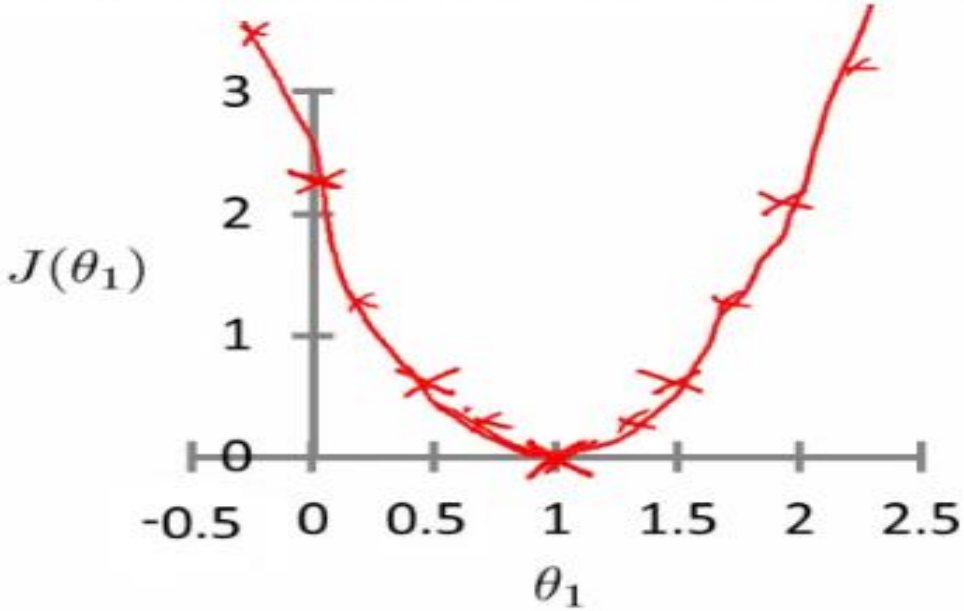


Theta 1 =0, J(theta1)= 2.3

Simple Linear Regression: Cost Function

If we compute a range of values plot

- $J(\theta_1)$ vs θ_1 we get a polynomial (looks like a quadratic)



Our aim is to find the value of θ_1 for which the value of $J(\theta_1)$ is minimum.

Simple Linear Regression: Cost Function

Now, we will see the original hypothesis and corresponding cost function.

Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

Parameters: θ_0, θ_1

Cost Function: $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$

Goal: $\underset{\theta_0, \theta_1}{\text{minimize}}\ J(\theta_0, \theta_1)$